

CURRICULUM VITAE

Stefano Casirati, Ph.D.

Postdoctoral Researcher

Department of Environmental Science, Policy, and Management
University Of California, Berkeley

Email: scasirati@berkeley.edu
Office: 222 Mulford Hall

EDUCATION

- | | | |
|------|--|--|
| 2023 | Ph.D. in Environmental Systems | University of California, Merced, California, USA
School of Engineering,
Advisors: Dr. Martha Conklin, Dr. Safeeq Khan |
| 2014 | M.Eng. Environmental Engineering, | Politecnico of Milan, Italy
Advisors: Dr. Maria Cristina Rulli, Dr. Paolo d'Odorico |
| 2011 | B.Eng. Environmental Engineering, | Politecnico of Milan, Italy
Advisor: Dr. Maria Cristina Rulli |

APPOINTMENTS

12/04/2023 - Present Postdoctoral Researcher, University of California, Berkeley, USA

ADDITIONAL PROFESSIONAL EXPERIENCES

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| 2015–2018 | Environmental engineer, energy efficiency analyst Suntrading S.p.A. / Sunsaving S.r.l. / Solarfast S.p.A., Italy |
| 2015 | Environmental engineer (HSE) Tecnoambiente S.r.l., Italy |
| 2014–2015 | Research internship: “ <i>Virtual water analysis for food and energy</i> ”

Project: “ <i>The role of Europe in land deals: understanding the food-energy-water security nexus</i> ” (PI Dr. Maria Cristina Rulli)

Department of Civil and Environmental Engineering, Politecnico of Milan, Italy |

PUBLICATIONS

Peer-Reviewed Journal Articles

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| 2026 | Casirati, S. , M. H. Conklin, and M. Safeeq. 2026. “Hydrological Response to Compounding Impacts of Climate Change and Forest Management in the Upper Kings River Basin, CA, USA.” <i>Ecohydrology</i> 19, no. 1: e70157. https://doi.org/10.1002/eco.70157 . |
| 2025 | Casirati, S. , D’Odorico, P., Reichle, R. H., Rouf, T., Stookey, M., & Giroto, M. (2025). A model-based evaluation of the effects of irrigation expansion on regional and global land surface climate. <i>Earth's Future</i> , 13, e2025EF006271. https://doi.org/10.1029/2025EF006271 |

- Casirati, S.,** Conklin, M., Nandi, S. and Safeeq, M. (2025), Effect of Forest Management Practices on Water Balance Across a Water–Energy Gradient in the Upper Kings River Basin, USA. *Ecohydrology*, 18: e2753. <https://doi.org/10.1002/eco.2753>
- 2023 **Casirati S.,** Conklin M.H. and Safeeq M. (2023) Influence of snowpack on forest water stress in the Sierra Nevada. *Front. For. Glob. Change* 6:1181819. doi: 10.3389/fgc.2023.118181
- 2019 Rulli M.C., **Casirati S.,** Dell’Angelo J., Davis K.F., Passera C., D’Odorico P. (2019) Interdependencies and telecoupling of oil palm expansion at the expense of Indonesian rainforest. *Renewable and Sustainable Energy Reviews* 105, 499-512. <https://doi.org/10.1016/j.rser.2018.12.050>

Conference Proceedings

- 2015 D’Odorico P., Rulli M.C., **Casirati S.,** Davis K.F., Dell’Angelo J., Land use change induced by large scale land acquisitions. World Bank Conference on Land and Poverty, Washington DC, 23-27 March 2015.

PRESENTATIONS

- 2026 **Casirati S.,** D’Odorico P., Fung I., Girotto M., (2026) “Irrigation and Land–Atmosphere Interactions: Insights into Global Climate Patterns”, Oral Presentation, 106th AMS American Meteorological Society Annual Meeting, Houston, USA
- 2025 **Casirati S.,** D’Odorico P., Fung I., Girotto M., (2025) “Irrigation's Global and Regional Climate Impacts: Exploring Land–Atmosphere Feedbacks Across S2S Timescales”, Poster Presentation, Advancing Understanding of Land-Atmosphere Interactions and Processes on S2S Predictability Workshop, Boulder, Colorado, USA
- Safeeq, M. and **Casirati, S.,** (2025) “Forest Management in a Warming World: Enhancing Insights into Compounding Hydrologic Impacts”, Oral Presentation, European Geophysical Union General Assembly 2025, Vienna, Austria.
- 2024 **Casirati S.,** D’Odorico P., Girotto M., (2024) “Evaluating the effects of irrigation expansion on regional and global land surface climate using a Land Surface Model”, Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
- Safeeq, M., **Casirati, S.,** Nandi, S. and Conklin, M., (2024) “Modeling the effects of forest thinning on the long-term water balance in a Mediterranean mixed-conifer basin”, Oral Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA
- 2022 **Casirati S.,** Safeeq M., and Conklin M. H., (2022) “Predicting drought-related tree mortality in the Sierra Nevada.”, eLightning presentation Frontiers in Hydrology Meeting – San Juan, Puerto Rico, USA
- 2020 **Casirati S.,** Safeeq M., and Conklin M. H., (2020) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Oral Presentation, Yosemite Hydroclimate Meeting, Virtual.
- 2020 **Casirati S.,** Safeeq M., and Conklin M. H., (2020) “Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada”, Poster Presentation, Sierra Nevada Research Institute - UC Merced Research Week 2020: Research Contributions: “Addressing

Climate Change for the Region and the World" - University of California Merced, California, USA

Casirati S., Safeeq M., and Conklin M. H., (2020) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Poster Presentation, Grad Visitation Poster Session: University of California Merced, California, USA

2019 **Casirati S.**, Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Poster Presentation, American Geophysical Union, Fall Meeting. San Francisco, California, USA

Casirati S., Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Oral Presentation, Headwaters to Groundwater Annual Meeting, University of California Davis, California, USA

Casirati S., Safeeq M., and Conklin M. H., (2019) "Influence of snow accumulation and ephemerality on recent tree mortality in the Sierra Nevada", Oral Presentation, Symposium: "Climate change and the ecology of Sierra Nevada forests", University of California Merced, California, USA

AWARDS and FELLOWSHIPS

2023 UC Merced ES Summer Fellowship (\$7500)
2022 AGU FIHM Travel Grant (\$1000)
2022 UC Merced ES Summer Fellowship (\$7000)
2019 Fall-1819 UC Merced ES Bobcat Travel Fellowship (\$1500)

TEACHING

University of California, Merced

2023 Field Methods in Snow Hydrology - ENVE 181 (*Teaching Assistant*)
2022 Surveying and Geomatics - CE010 (*Teaching Assistant*)
2022 Field methods in Snow Hydrology - ENVE 181 (*Teaching Assistant*)
2022 Calculus I - MATH 011 (*Teaching Assistant*)
2020 Spatial Analysis & Modeling – ENGR 180 (*Guest Lecture*)

MENTORING

2025- Chloe Faehndrich (UC Berkeley, Ph.D)
2024- Sarah Ryu (UC Berkeley, BS)
2020 Breanna Xiong (UC Davis, BS *Summer Intern California Ecosystem Climate Solutions project*)

ACADEMIC and PUBLIC SERVICE

2025 Panelist, NASA Grant Proposal Reviewer. Future Investigators in NASA Earth and Space Science and Technology (FINESST) Earth Panelist (Spring 2025)

- 2022 Equipment removal for the Hemlock Forest Restoration Project, Sierra Nevada Research Institute
- 2021 Environmental Systems seminar graduate student lead
- 2020 Environmental Systems seminar graduate student lead
- 2019 Weir repair activities at the Kings River Experimental Watershed, US Forest Service
- Ongoing Reviewer: *Journal of Hydrology*, *Journal of Advances in Modeling Earth Systems*, *Journal of Hydrometeorology*, *Theoretical and Applied Climatology*

ADDITIONAL COURSES

- 2021 Principle of Pedagogy course
- 2020 Processed-Based Hydrological modeling – University of Saskatchewan - Centre for Hydrology
- 2019 CUAHSI Master Class: Advanced Techniques in Watershed Science - University of Arizona - Biosphere 2

RESEARCH INTERESTS

Eco-Hydrological Modeling, Computational Hydrology, Remote Sensing, Forests, Agriculture, Irrigation, Earth System Models, Food Energy Water Systems, Land-Atmosphere Interactions, Climate Teleconnections.

PROGRAMMING AND COMPUTING SKILLS

R, Python, ArcGIS, QGIS, CropWat, SWAT+, SUMMA, Google Earth Engine, NASA GEOS, NASA CLSM